

One Interval, Two Cutoffs

Unit 4 · Topic 4.3 · TODAY'S PROBLEM

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE (5 min, on your sheet)

Which approach is more defensible? Cite an endpoint or identify the quantity being estimated.

- DOK 1** (a) Define the population mean and locate 42 and 45 relative to the interval.
- DOK 2** (b) Verify random, 10 percent, and shape conditions. Reproduce the standard error, margin of error, and interval; interpret it in context.
- DOK 3** (c) Adjudicate the approaches. Use both endpoints to evaluate the two less-than claims, explain why the cutoff locations matter, distinguish the population mean from individual times, and trace what the rejected interpretation could lead the library to advertise.

Video + follow-along: [u7_lesson3_live.html](https://www.khanacademy.org/a/u7_lesson3_live.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

